



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

OscarCarranzaFeynman
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Outline

Executive summary and project question

Data collection, wrangling and analytical methods

Visual EDA and SQL results

Folium maps and Plotly Dash dashboard

Classification results, limitations and conclusions

Executive Summary

Question: can historical launch characteristics predict a Falcon 9 landing outcome?

90-launch model sample: 60 successes, a 66.67% success rate.

KSC LC-39A leads the 56-launch dashboard sample: 10/13 successes (76.9%).

All four classifiers achieve 83.33% test accuracy (15/18). The decision tree has the highest cross-validation score (87.68%).

Use these results as exploratory evidence; the small test sample and preprocessing leakage limit deployment confidence.

Introduction

Reusable first stages can reduce launch costs; predicting landing outcomes supports assessment of reusability.

Explore how launch site, orbit, payload and flight history relate to outcomes.

Combine data preparation, SQL, charts, geographic analysis and classification into one reproducible workflow.

Target: the lab's binary landing outcome label, not mission delivery success or a directly measured cost estimate.

Section 1

Methodology

Methodology

1. Inspect IBM-provided historical launch data and its upstream sources.
2. Count sites, orbits and outcomes; create the binary Class label.
3. Explore relationships with scatter plots, rates and SQL aggregates.
4. Build interactive Folium maps and a Dash dashboard.
5. Encode 80 features; standardize, split and tune four classifiers.
6. Compare validation scores, test accuracy and confusion matrices.

Data Collection

Three source snapshots serve different analyses:

Wrangling / visualization / ML: 90 launches, 2010–2020.

SQL: 101 launch records, with mission and landing outcomes stored separately.

Folium / Dash: 56 launches and four site labels.

IBM Skills Network supplies CSV snapshots derived from launch records. Counts and rates must be interpreted within their own snapshot.

Repository: <https://github.com/OscarCarranzaFeynman/spacex-data-science-capstone>

Data Collection — SpaceX API

Upstream lab workflow:

GET launch records → filter Falcon 9 → resolve payload, rocket, core and launchpad references → flatten fields → export a CSV.

Fields used downstream include payload mass, orbit, site coordinates, core serial and landing outcome.

This report verifies the supplied snapshots and downstream notebooks. A separately executed API collection notebook was not provided, so a fresh API collection is not claimed.

Data Collection — Web Scraping

Upstream lab workflow:

Request the historical launch table → parse HTML tables → normalize dates, booster versions, payloads and site names → export structured records.

The SQL snapshot retains mission success and landing success as different columns.

A separately executed scraping notebook was not provided. SQL results in this report come from the IBM-supplied launch table loaded into SQLite.

Data Wrangling

Count LaunchSite, Orbit and Outcome; cross-tabulate outcomes by orbit.

Class = 0 for False ASDS, False Ocean, False RTLS, None ASDS or None None; otherwise Class = 1.

The 90-row sample contains 60 Class=1 and 30 Class=0. True ASDS appears 41 times; True RTLS 14 times; True Ocean 5 times.

LandingPad is missing in 28.89% of rows; no landing pad can be legitimate. Payload mass uses the supplied imputed value.

Export dataset_part_2.csv, then one-hot encode Orbit, LaunchSite, LandingPad and Serial: 80 numeric features.

EDA with Data Visualization

Scatter plots: flight number / site, payload / site, flight number / orbit and payload / orbit, colored by Class.

Orbit bar chart: mean Class estimates the landing success rate within each orbit.

Yearly line chart: group by launch year and average Class.

Interpret patterns alongside sample counts and changing launch technology; avoid treating association as causation.

EDA with SQL

Load the SQL extension and connect to SQLite using %sql.

Load SPACEXTBL (101 rows) and execute queries with %%sql.

DISTINCT / LIKE / LIMIT identify sites and example launches.

SUM / AVG / MIN summarize payload and first ground landing.

WHERE filters drone landings, dates and payload ranges.

GROUP BY / COUNT / ORDER BY summarize outcomes and rank frequencies.

Interactive Maps with Folium

Mark four site labels using circles and named markers.

MarkerCluster groups overlapping launches; green markers show Class=1 and red markers Class=0.

Add proximity markers and lines for the coastline, railway, road, highway and city.

Haversine distance uses latitude/longitude and $R=6,373$ km. Distances refer to selected nearby points, not exhaustive nearest-feature searches.

Dashboard with Plotly Dash

Site dropdown offers ALL and four launch-site labels.

ALL-site pie: count successful launches per site. Individual-site pie: success versus failure counts.

Payload RangeSlider spans 0–10,000 kg with 1,000 kg steps.

A callback filters the scatter plot by both site and payload range. Points show payload mass versus Class, colored by booster version.

The interactions expose mission mix and sample composition without changing the underlying data.

Predictive Analysis

Target Y: Class. Features X: 90 rows × 80 encoded columns.

Follow the lab: StandardScaler before train_test_split (random_state=2, test_size=0.2): 72 train / 18 test.

GridSearchCV with 10 folds tunes logistic regression, SVM, decision tree and KNN.

Select parameters using validation scores; evaluate held-out accuracy and confusion matrices.

Limitation: fitting the scaler before splitting leaks test information. A future Pipeline must fit scaling only inside training folds.

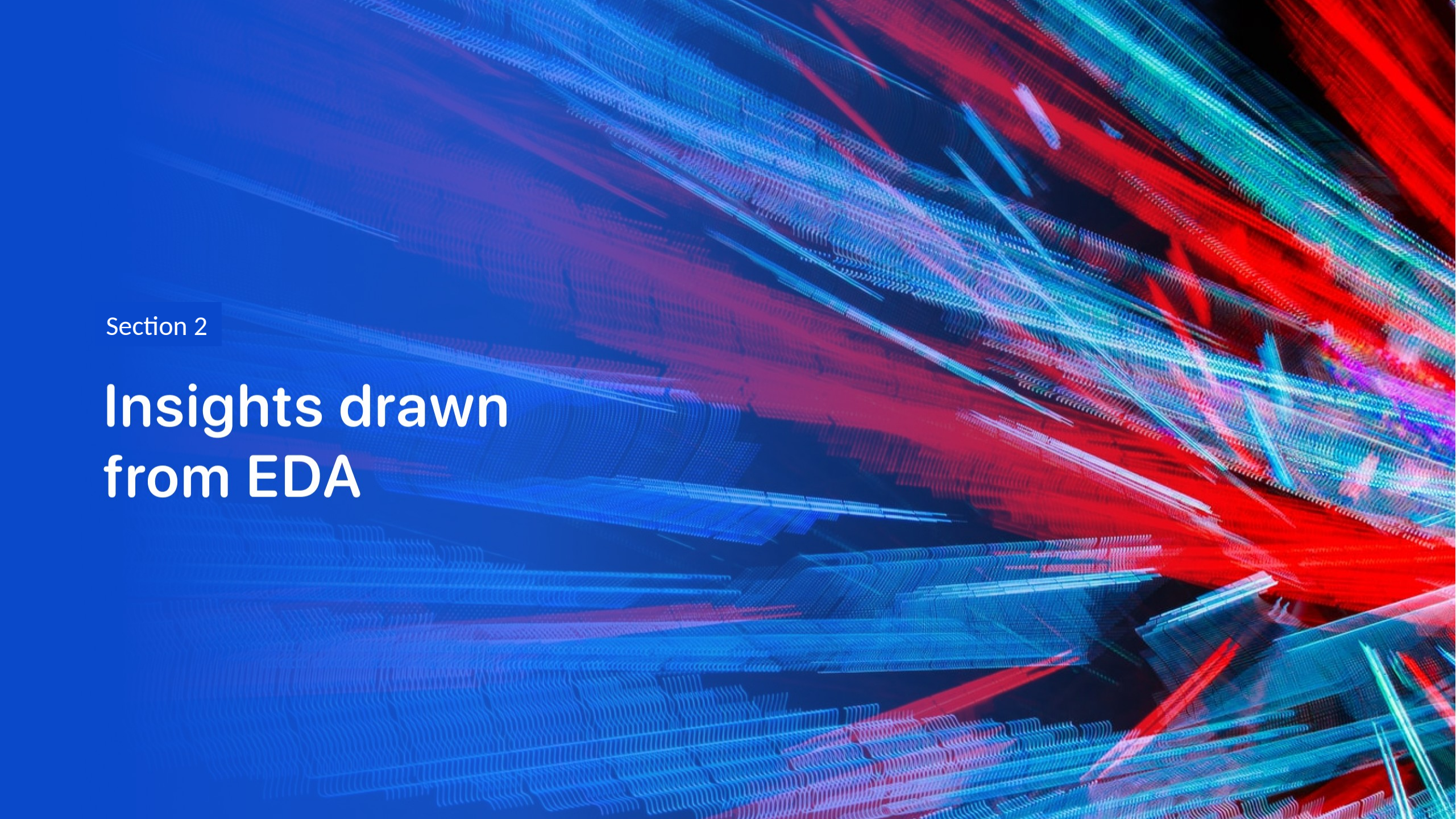
Results

Visual EDA: rates generally improve with launch history, with a dip in 2018. Orbit and payload patterns vary by mission mix.

SQL: distinguish mission delivery from first-stage landing; identify historical milestones and payload totals.

Maps / Dash: coastal sites, nearby infrastructure and site-specific outcome patterns.

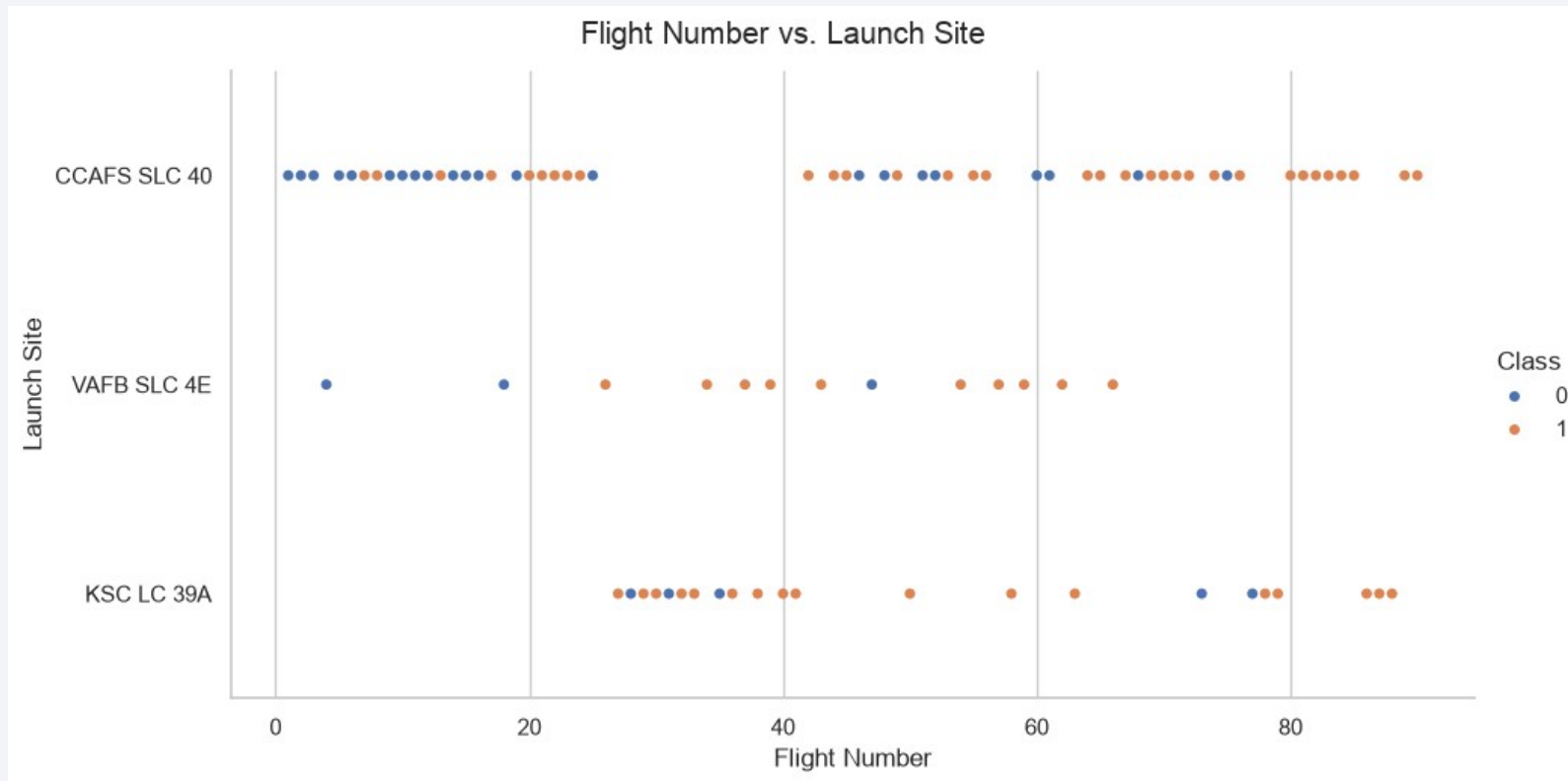
Classification: identical 83.33% test accuracy across four models; decision tree leads cross-validation in this run.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

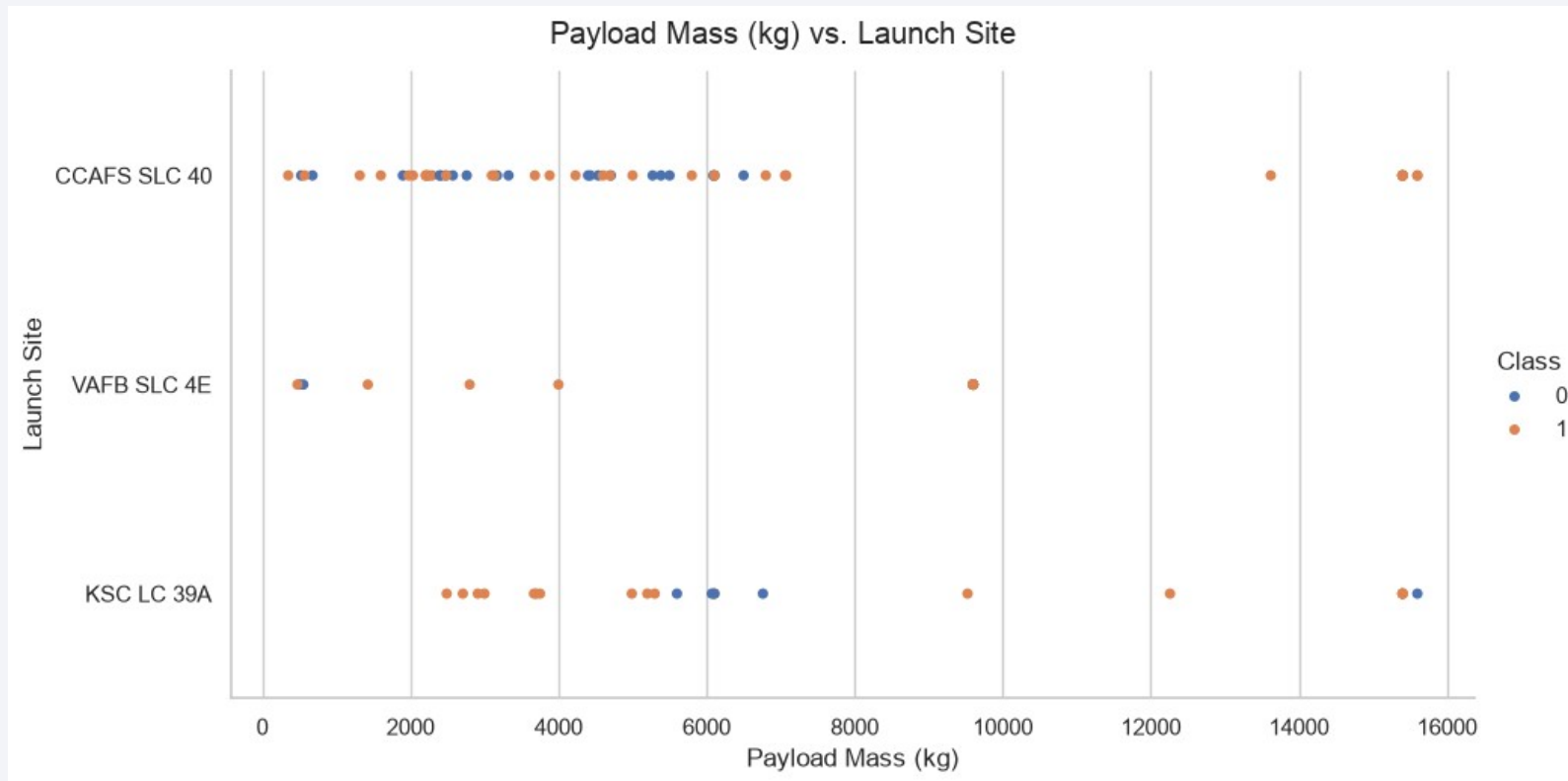


Later flights contain more successful outcomes.

Site comparisons also reflect time and mission mix.

90-launch snapshot.

Payload vs. Launch Site

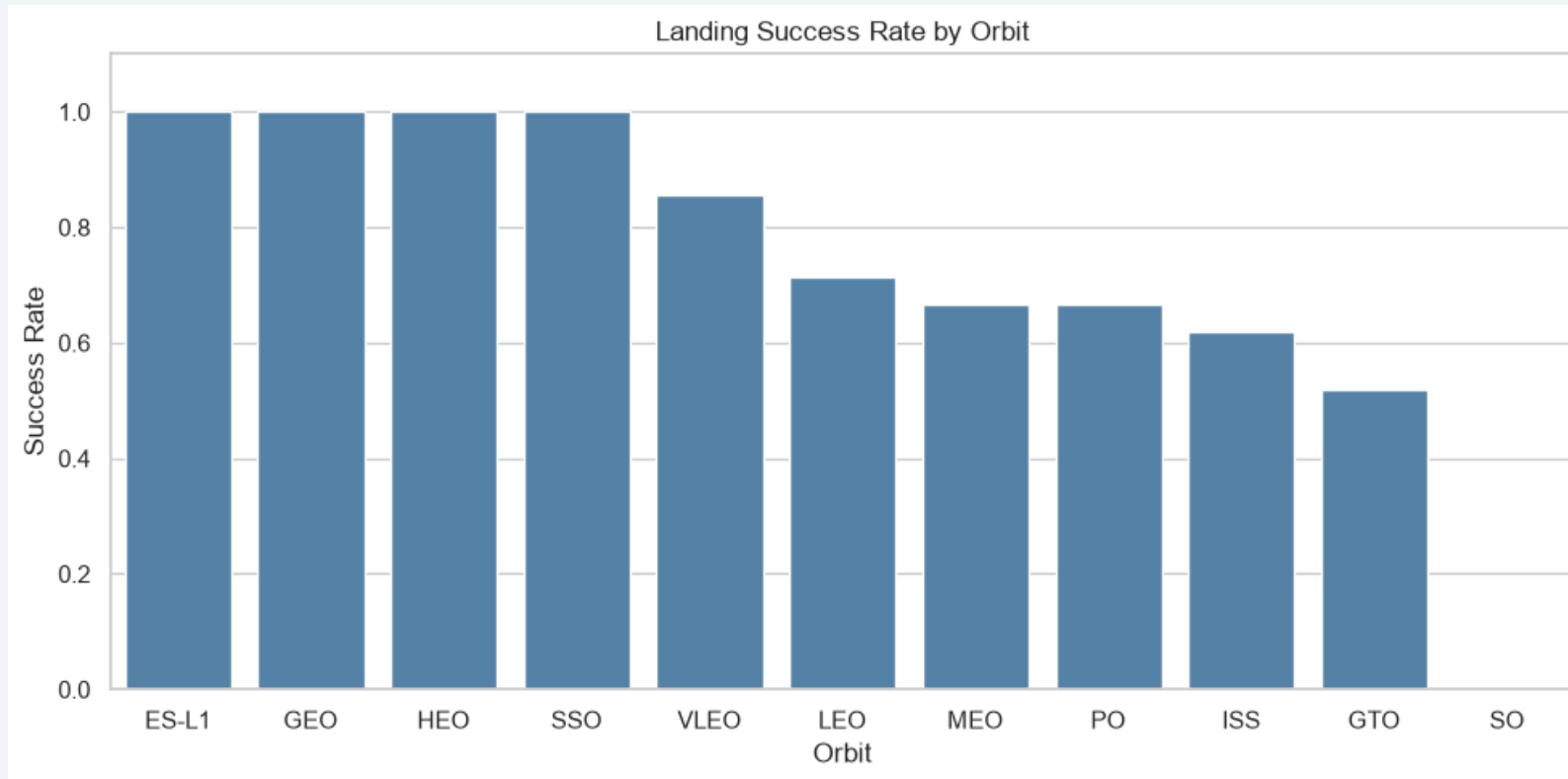


Payload ranges differ across launch sites.

Both successful and failed labels appear at several payload levels.

Payload alone does not explain outcomes.

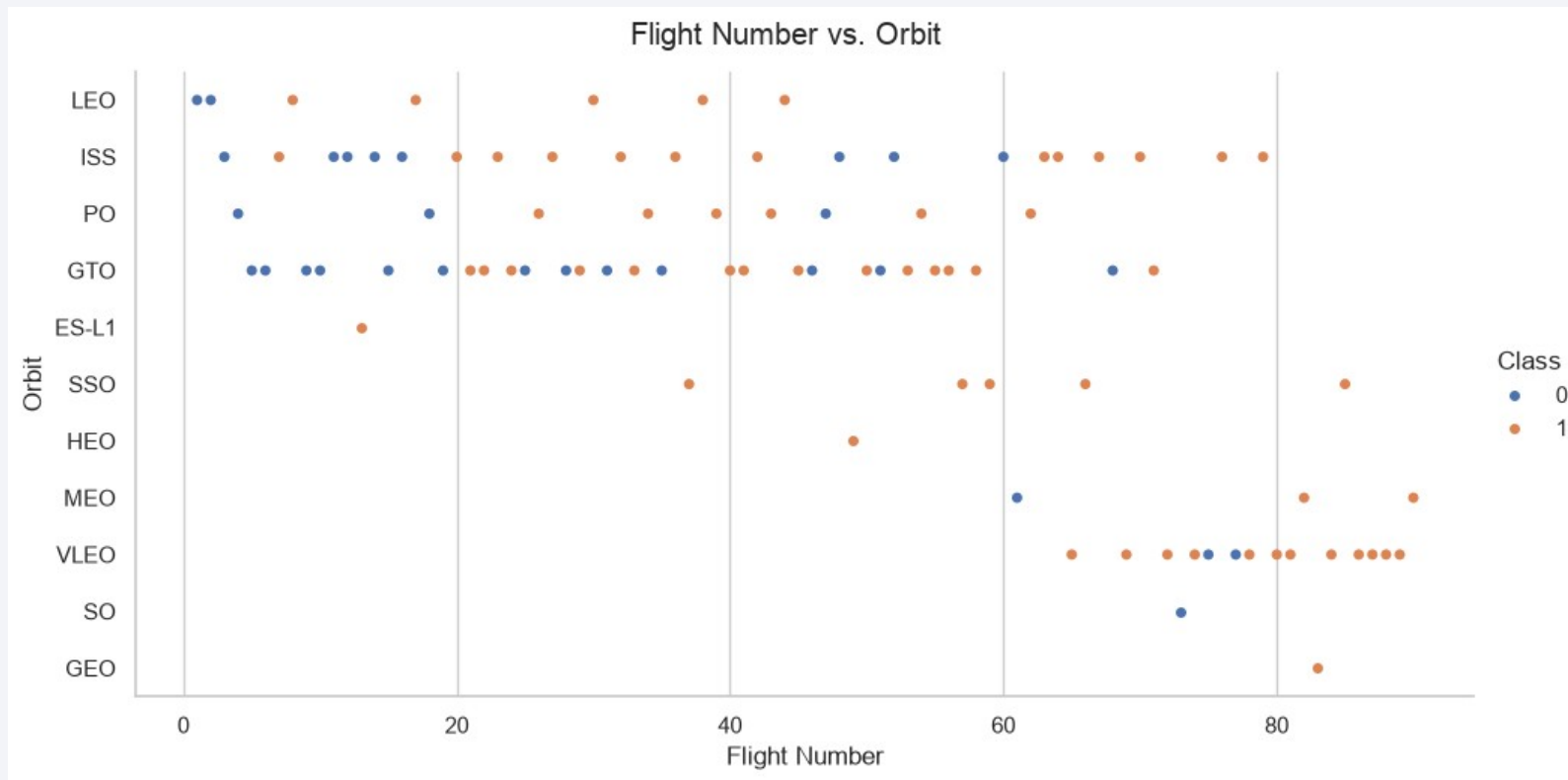
Success Rate vs. Orbit Type



SSO: 100% (5/5).
VLEO: 85.7% (12/14).
GTO: 51.9% (14/27).

Several 100% orbit rates use only one launch; they are weak evidence.

Flight Number vs. Orbit Type

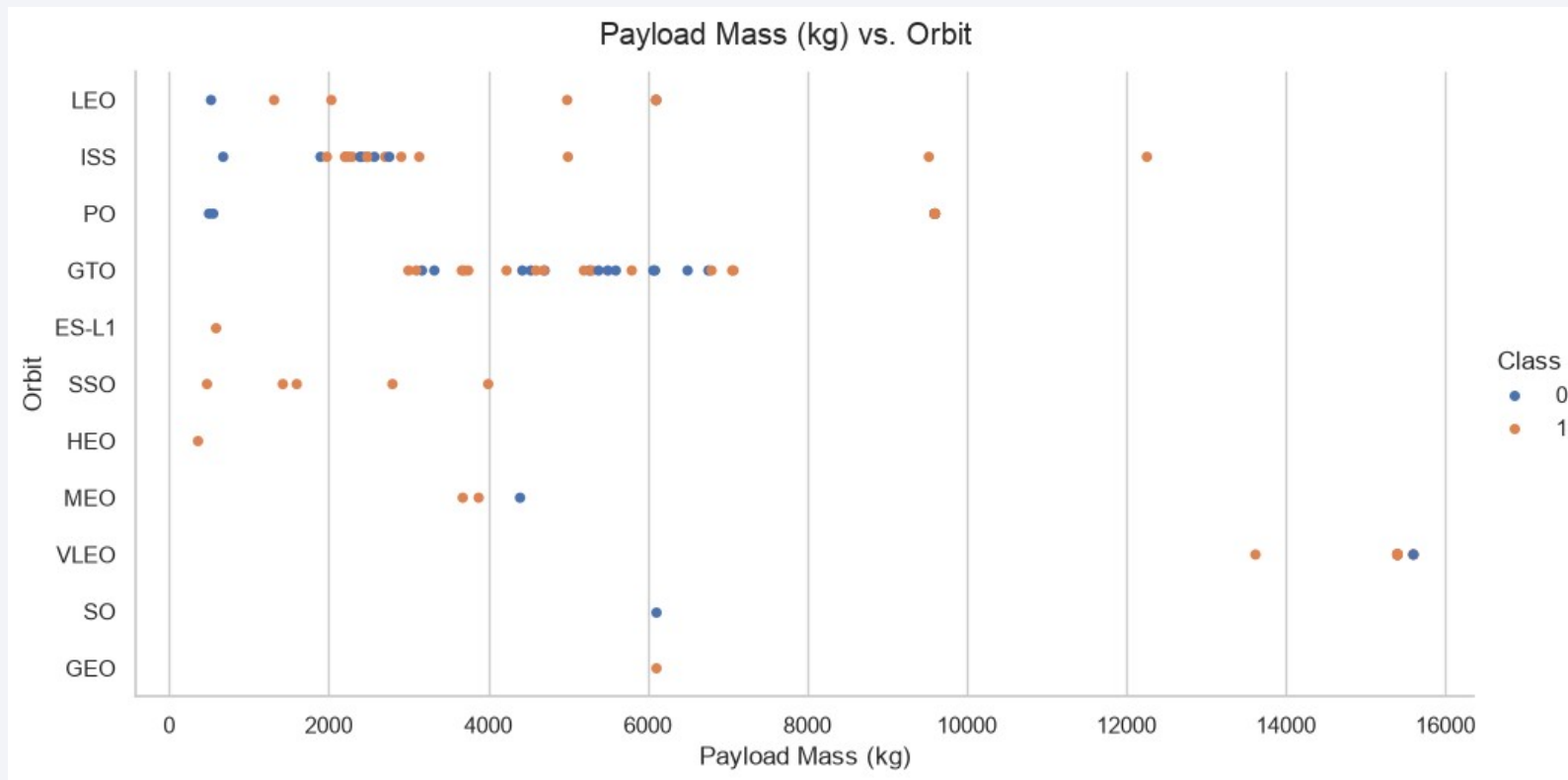


Orbit categories occur in different periods.

Flight history and orbit are related to mission selection and technology changes.

Compare within time periods before attributing effects.

Payload vs. Orbit Type

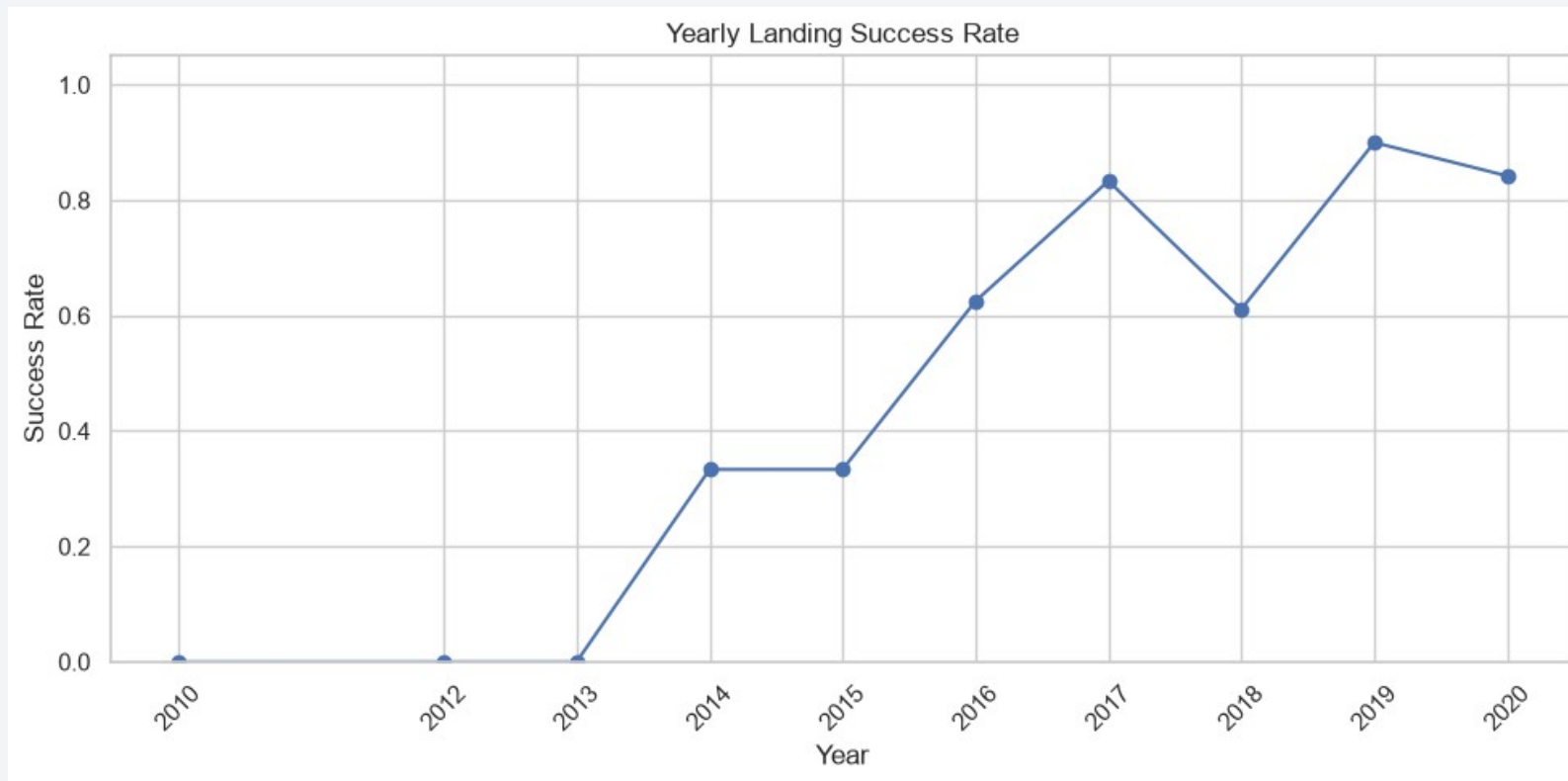


Payload and orbit are connected by mission objectives.

GTO spans a broad payload range.

Sparse categories prevent confident ranking from a few observations.

Launch Success Yearly Trend



2016: 62.5%

2017: 83.3%

2018: 61.1%

2019: 90.0%

2020: 84.2%

Long-term improvement is not monotonic.

All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Four labels in this snapshot; LC-40 and SLC-40 naming varies historically.

Launch Sites Beginning with CCA

```
SELECT Date, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5
```

Date	Booster_Version	Launch_Site
2010-06-04	F9 v1.0 B0003	CCAFS LC-40
2010-12-08	F9 v1.0 B0004	CCAFS LC-40
2012-05-22	F9 v1.0 B0005	CCAFS LC-40
2012-10-08	F9 v1.0 B0006	CCAFS LC-40
2013-03-01	F9 v1.0 B0007	CCAFS LC-40

Total Payload Mass

NASA (CRS) payload total: 45,596 kg

```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)';
```

This is the total for the specifically named customer group in the SQL snapshot. It is not the total payload for every NASA-related mission.

Average Payload Mass — F9 v1.1

Average payload: 2,928.4 kg

```
SELECT AVG(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Booster_Version LIKE 'F9 v1.1%';
```

The filter matches the version prefix while retaining individual booster identifiers.

First Successful Ground Landing

First successful ground-pad landing: 22 December 2015

```
SELECT MIN(Date)
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

The earliest date is a historical milestone within the supplied SQL records.

Drone Landing: Payload 4–6 Tonnes

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE Landing_Outcome='Success (drone ship)' AND  
PAYLOAD_MASS__KG_ >4000 AND PAYLOAD_MASS__KG_ <6000
```

Booster_Version	PAYLOAD_MASS__KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

Mission Success versus Landing Success

SQL mission outcomes: 100 successful missions / 1 failed mission.

Mission_Outcome values beginning with “Success” are grouped together, including qualified success labels.

Mission success concerns payload delivery. Landing success concerns first-stage recovery.

The 99.01% mission success share in the SQL sample is therefore different from the 66.67% binary landing-label rate in the 90-launch model sample.

Boosters with Maximum Payload

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_)
FROM SPACEXTBL)
```

Booster version	Payload (kg)	Booster version	Payload (kg)
F9 B5 B1048.4	15 600	F9 B5 B1049.5	15 600
F9 B5 B1049.4	15 600	F9 B5 B1060.2	15 600
F9 B5 B1051.3	15 600	F9 B5 B1058.3	15 600
F9 B5 B1056.4	15 600	F9 B5 B1051.6	15 600
F9 B5 B1048.5	15 600	F9 B5 B1060.3	15 600
F9 B5 B1051.4	15 600	F9 B5 B1049.7	15 600

Failed Drone Landings in 2015

SELECT Date,Booster_Version,Launch_Site FROM SPACEXTBL WHERE Date LIKE '2015%' AND Landing_Outcome='Failure (drone ship)'

Date	Booster_Version	Launch_Site
2015-01-10	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	F9 v1.1 B1015	CCAFS LC-40

Landing Outcomes: 2010–2017

```
SELECT Landing_Outcome,COUNT(*) AS Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY  
Landing_Outcome ORDER BY Count DESC
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

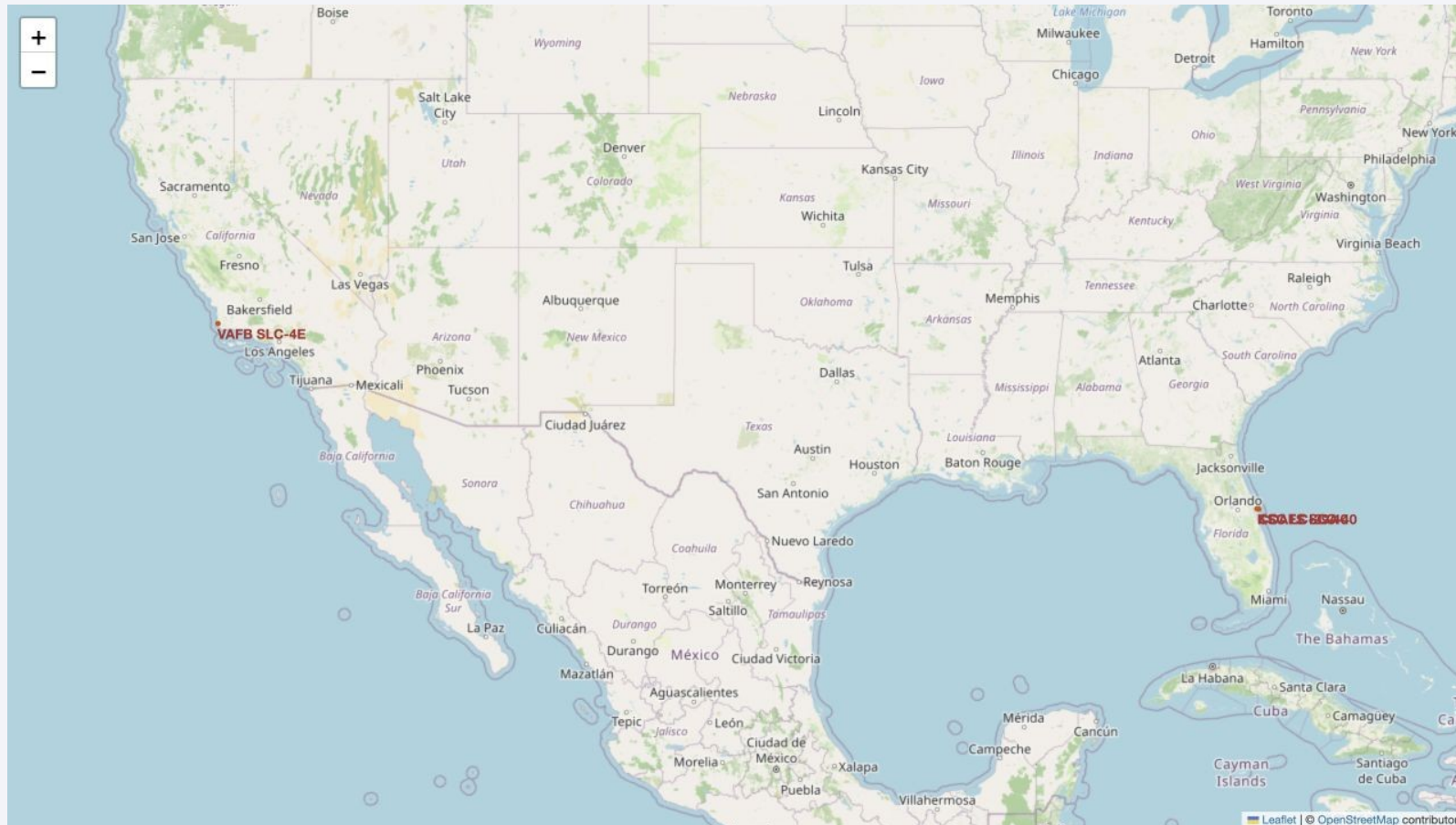
31 records in the inclusive date range. No attempt is the largest group.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

Launch Sites on the US Coasts

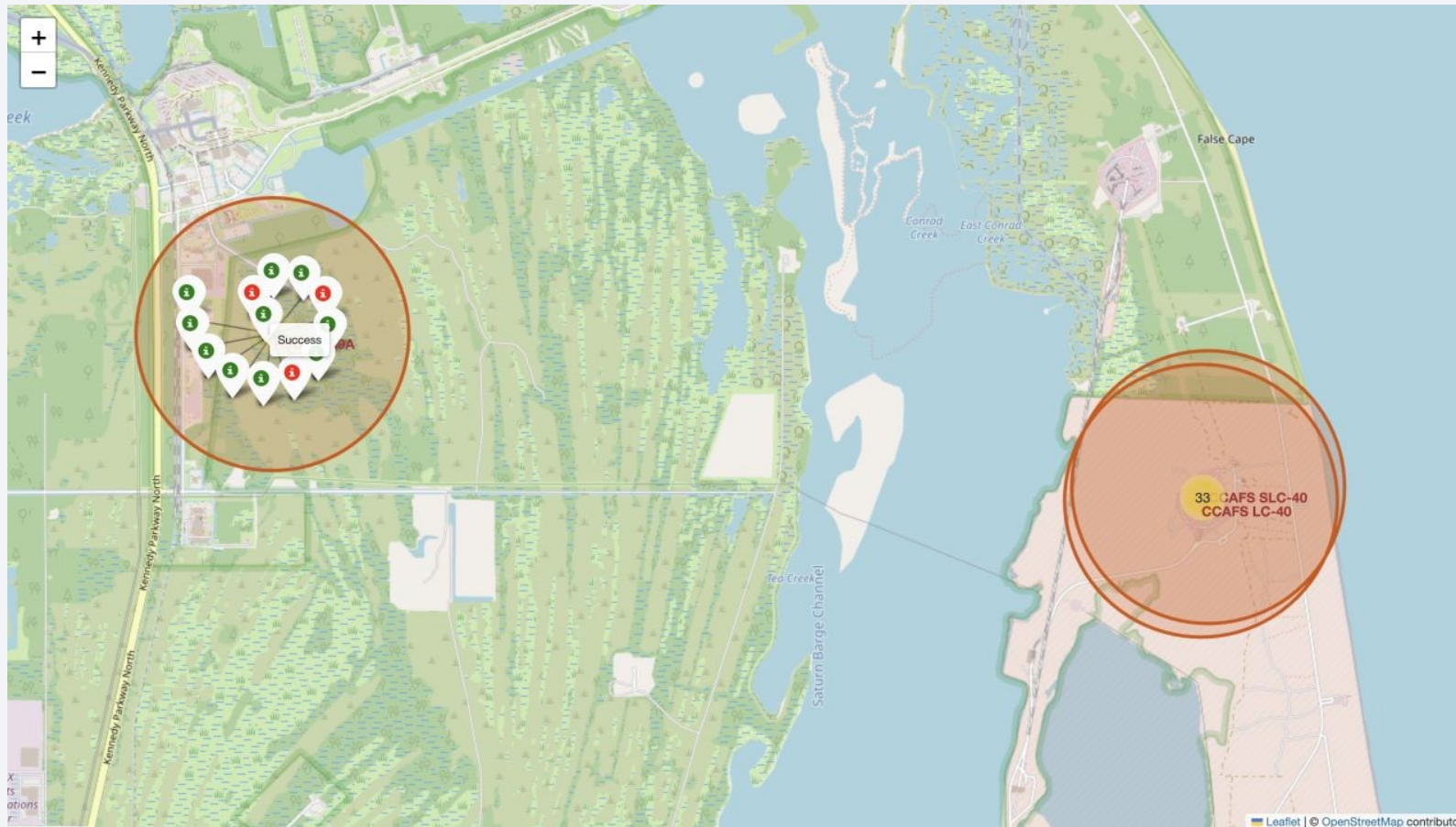


Four labeled sites appear on the Florida and California coasts.

Florida labels overlap at this zoom. The notebook allows zooming and marker inspection.

Map © OpenStreetMap contributors.

Clustered Success and Failure Markers

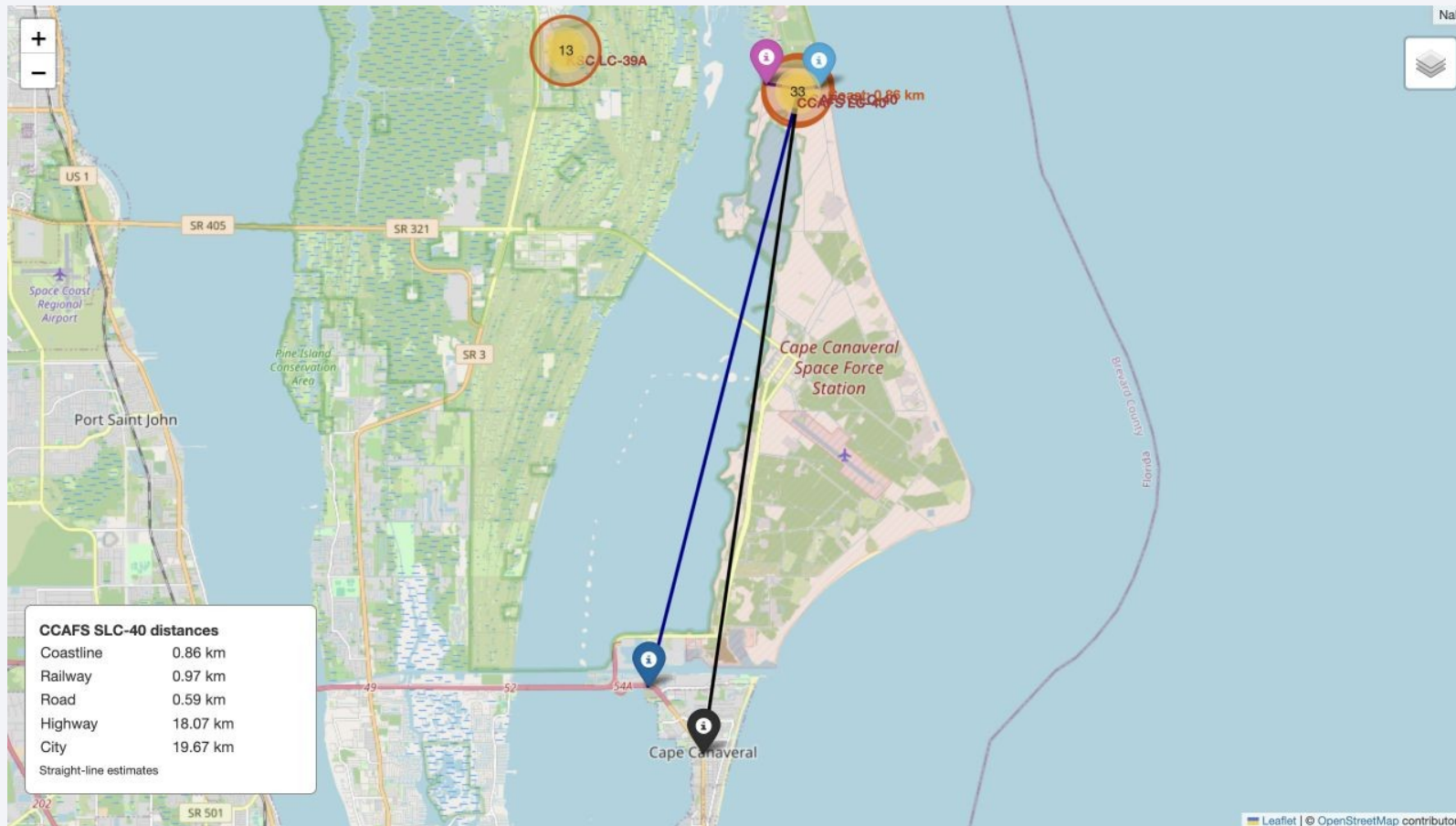


Expanded KSC cluster: 10 successes and 3 failures.

Green / red markers preserve individual outcomes despite shared site coordinates.

Map © OpenStreetMap contributors.

CCAFS SLC-40 Proximity Analysis



Selected straight-line distances:

Coast: 0.86 km

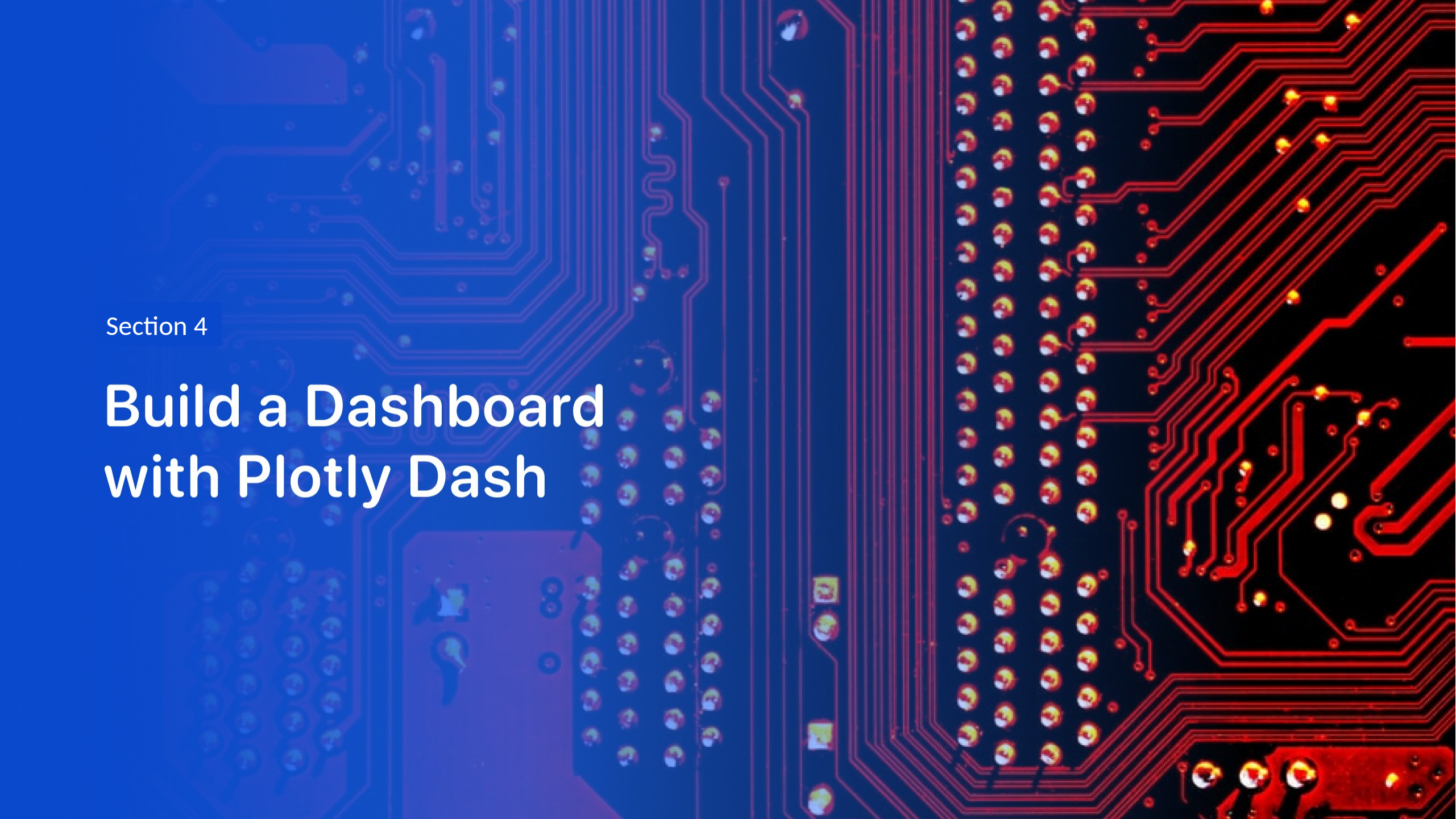
Railway: 0.97 km

Road: 0.59 km

Highway: 18.07 km

City: 19.67 km

OSM feature points; not travel distances.



Section 4

Build a Dashboard with Plotly Dash

Successful Launches by Site

SpaceX Launch Records Dashboard

Launch Site:

All Sites

Successful Launches by Site



Of 24 successful launches:

KSC LC-39A: 41.7%

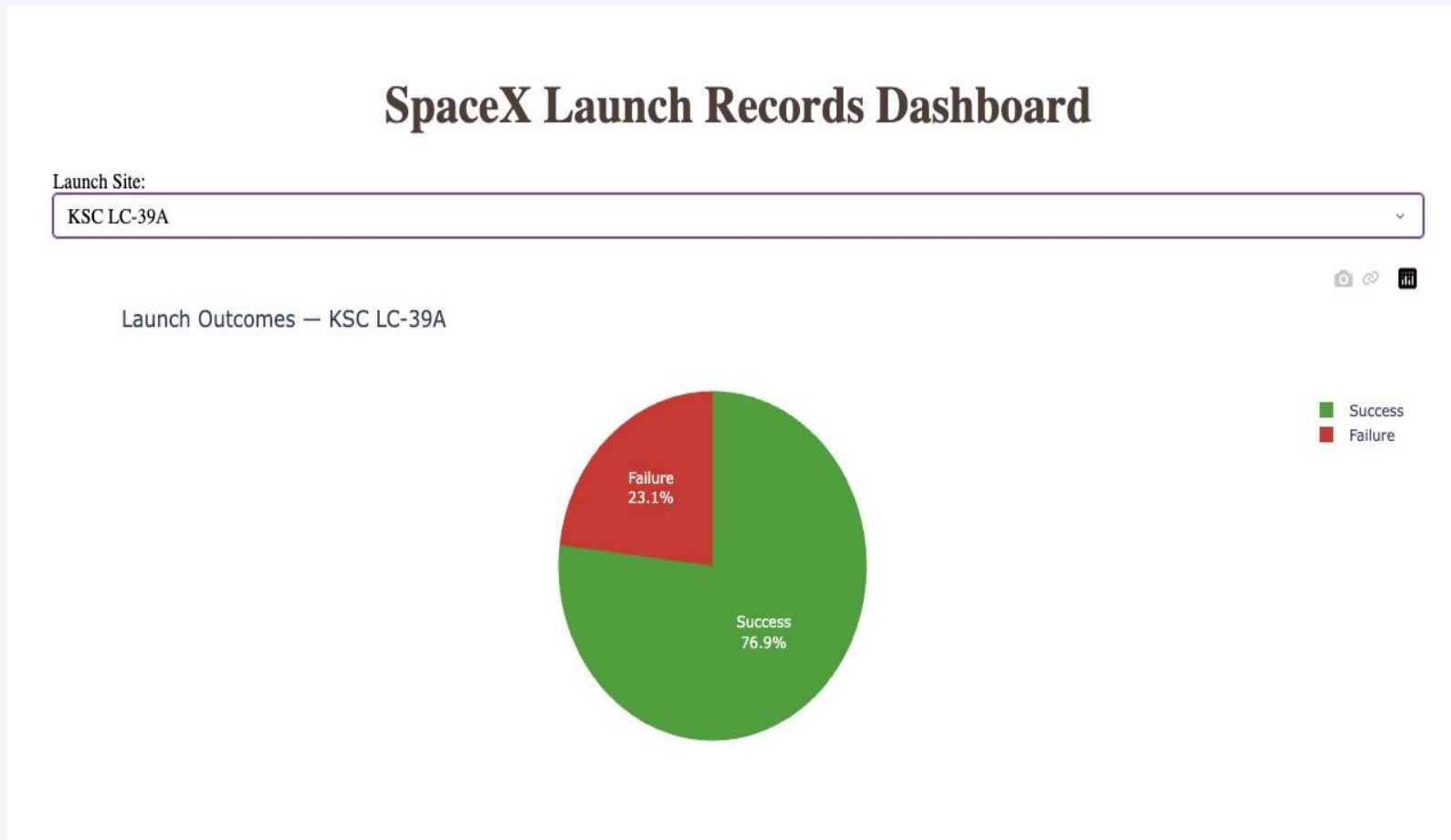
CCAFS LC-40: 29.2%

VAFB SLC-4E: 16.7%

CCAFS SLC-40: 12.5%

These are shares of successes,
not each site's success rate.

KSC LC-39A: Highest Success Rate

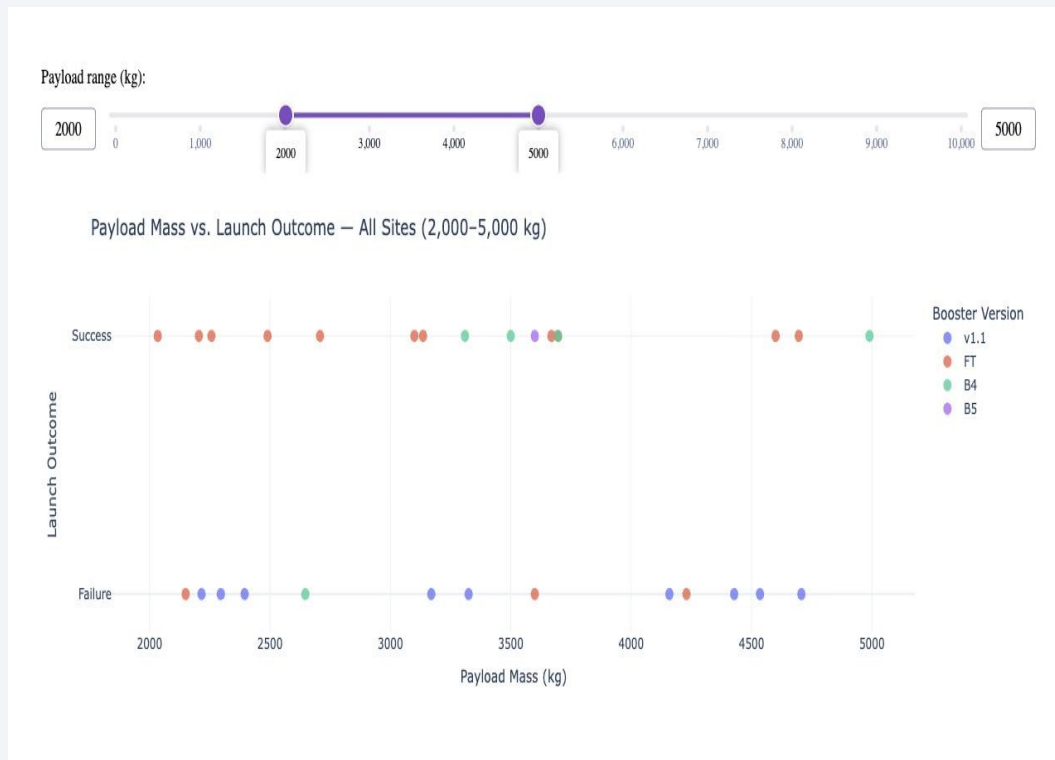


KSC LC-39A:
10 successes / 13 launches
Success rate: 76.9%

It leads this 56-launch sample
by both successful count and
rate.

Site advantage may reflect
mission and time differences.

Payload Filter: 2,000–5,000 kg



ALL sites (left) versus KSC LC-39A (right), using the same payload filter.

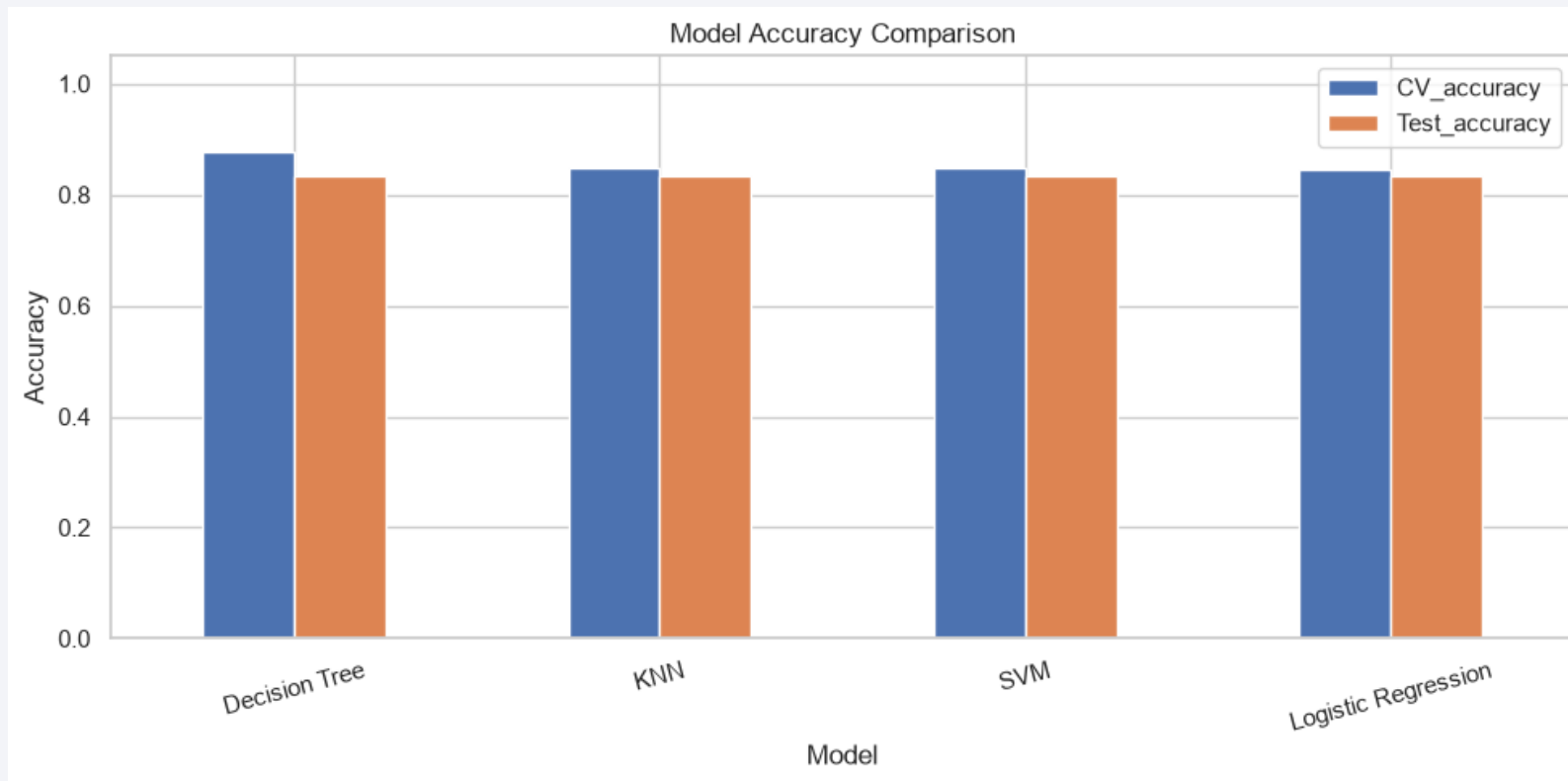
The filtered scatter shows both successful and failed outcomes; booster version and site alter the sample composition.



Section 5

Predictive Analysis (Classification)

Classification Accuracy



Test accuracy: 83.33% for every model (15/18).

CV scores:

Logistic: 84.64%

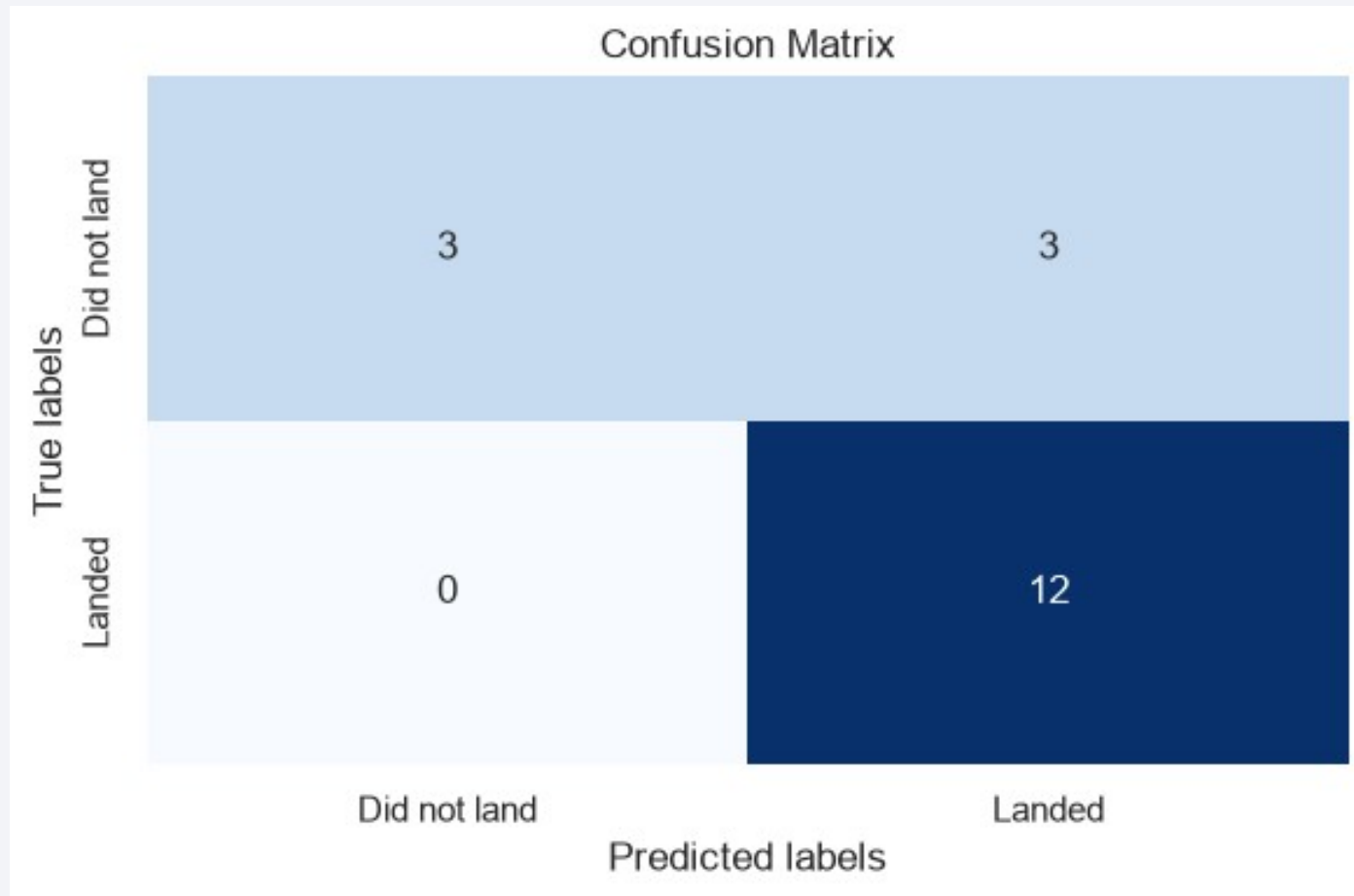
SVM: 84.82%

Tree: 87.68%

KNN: 84.82%

Tree leads validation; test scores are tied.

Decision Tree Confusion Matrix



TN=3, FP=3
FN=0, TP=12

Recall: 100%
Precision: 80%
Specificity: 50%
Balanced accuracy: 75%

Half of actual failures are
predicted successful.

Conclusions

Historical patterns support landing-outcome prediction, but the evaluation sample is small.

A 66.67% landing-label baseline and 83.33% test accuracy suggest useful signal; preprocessing leakage prevents a clean generalization claim.

KSC LC-39A leads the dashboard sample; do not infer a causal site effect without adjusting for year, orbit and booster version.

New insight: the 83.33% accuracy hides 50% specificity. Operational decisions should weight false-success predictions more heavily.

Next: use a chronological split, fold-local preprocessing, class-aware metrics and calibrated probabilities.

Appendix — Reproducibility and Limits

Best parameters:

Logistic: $C=0.01$, L2, lbfgs.

SVM: sigmoid, $C=1$, $\gamma=0.0316228$.

Tree: gini, depth=8, sqrt features, leaf=2, split=10.

KNN: 10 neighbors, Manhattan distance ($p=1$).

Public repository (completed notebooks and Python app):

<https://github.com/OscarCarranzaFeynman/spacex-data-science-capstone>

Sources: IBM Skills Network lab CSVs / presentation template; OpenStreetMap feature coordinates.

Assistance: ChatGPT.

All results correspond to these historical snapshots; no claim is made about current SpaceX operations.

Source: completed SpaceX notebooks; IBM Skills Network lab datasets.

Thank you!

SpaceX Falcon 9 Landing Prediction
Public notebooks, dashboard and report

